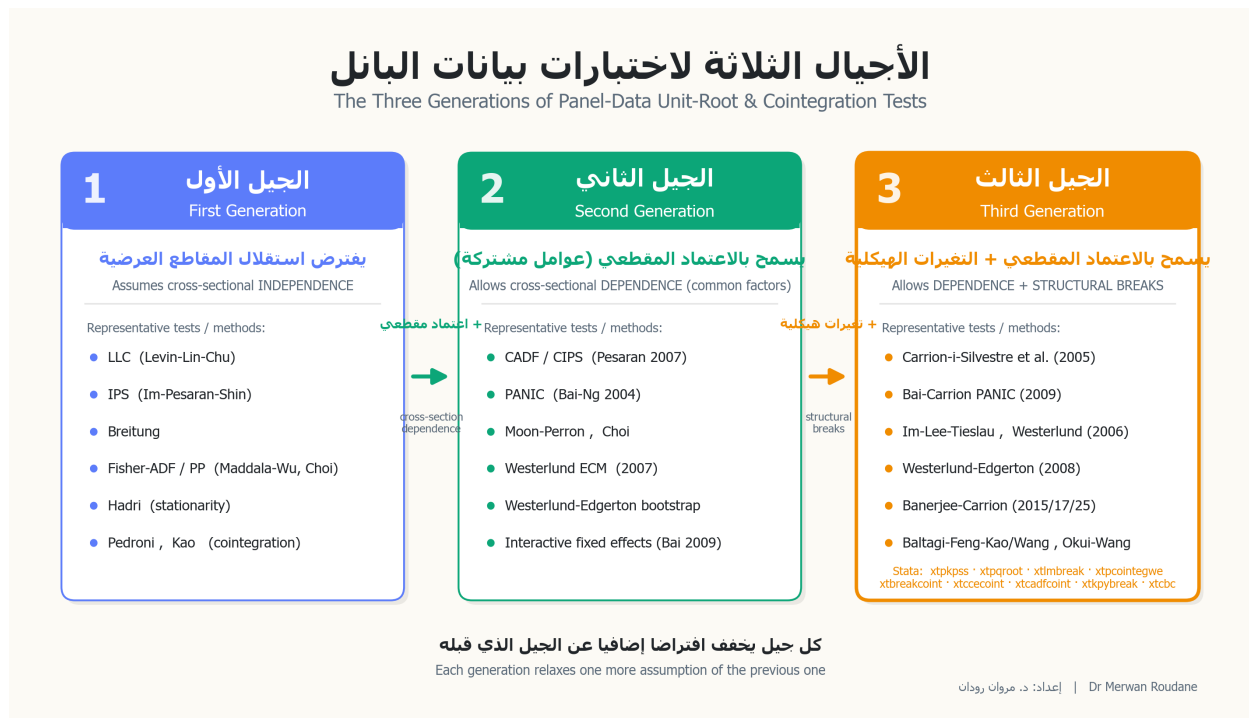


Third-Generation Panel Data Econometrics

Structural Breaks and Cross-Sectional Dependence

A Complete Lecture from Zero — for Beginners



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Contents

Notation and symbols	3
1 Introduction: what this lecture is about	3
2 Part I — Foundations from zero	3
2.1 The panel and its two dimensions	3
2.2 Stationarity, unit roots, and integration	4
2.2.1 Two families of test: ADF vs KPSS	4
2.3 Spurious regression and cointegration	5
2.4 Why “generations”?	5
3 Part II — Anatomy of a structural break	5
3.1 Decomposition: where a break lives	5
3.2 Sharp breaks: level, trend, regime	5
3.3 Where the break sits (four possibilities)	6
3.4 Common vs heterogeneous breaks (across units)	6
3.5 Vector vs coefficient-by-coefficient (across coefficients)	6
3.6 Sharp vs smooth breaks	6
3.7 A seven-question checklist	6
4 Part III — Cross-sectional dependence (CSD)	7
4.1 What CSD is and why first-generation tests fail	7
4.2 Two ways to remove CSD	7
4.2.1 (a) Cross-sectional averages — CCE (Pesaran, 2006)	7
4.2.2 (b) Common factors — PANIC / principal components (Bai–Ng, 2004)	7
4.3 The degree of dependence: classic binary vs modern	8
4.4 Detecting CSD: the CD test	8
5 Part IV — Types of tests: classic vs modern	8
5.1 Classic: existence (yes / no)	8
5.2 Confirmatory analysis	8
5.3 Modern: the degree, not just the fact	8
6 Part V — Three toolkits for modelling breaks	9
7 Part VI — Regularization explained, and GAGFL / PLS / CBC	9
7.1 What is regularization? (from zero)	9
7.1.1 Ridge, lasso, and why lasso hits zero	9
7.1.2 Fused lasso = automatic break detection	10
7.1.3 How it is computed: block coordinate descent (BCD)	10
7.2 PLS / AGFL — common breaks (Qian & Su, 2016)	10
7.3 GAGFL — grouped heterogeneous breaks (Okui & Wang, 2021)	10
7.4 CBC — coefficient-by-coefficient breaks (Kaddoura, 2025)	11
7.5 Breaks across quantiles (Zhang et al., 2022)	11
8 Part VII — Fourier methods in detail	11
8.1 The flexible Fourier form (FFF)	11
8.2 Fourier CCEMG (Guliyev, 2026)	12
8.3 Fourier panel ARDL / CS-ARDL	12

9 Part VIII — The third-generation tests, catalogued	12
9.1 Panel unit-root / stationarity with breaks	12
9.1.1 Panel KPSS with multiple breaks — Carrion et al. (2005) — <code>xtpkpss</code> . . .	12
9.1.2 PANIC with breaks — Bai & Carrion (2009)	13
9.1.3 Im–Lee–Tieslau panel LM (2005)	13
9.1.4 Quantile CIPS and tFR — <code>xtpqroot</code>	13
9.2 Panel cointegration with breaks	13
9.3 CCE with I(1) factors and breaks — <code>xtkpybreak</code>	13
9.4 Estimating and dating breaks: the shared machinery	13
9.4.1 Single break by SSR grid search	13
9.4.2 Multiple breaks	14
9.4.3 Why the estimated date behaves as if known	14
9.4.4 Pooling to a standard normal	14
10 Part IX — Practical workflow and Stata code	14
11 Part X — Each cointegration test in detail	15
11.1 Westerlund (2006) — LM test with multiple breaks (<code>xtlmbreak</code>)	15
11.2 Westerlund–Edgerton — factor and bootstrap versions	15
11.2.1 2008, factor-based (<code>xtpcointegwe</code>)	15
11.2.2 2007, bootstrap (<code>xtpcointegboot</code>)	15
11.3 The Banerjee–Carrion trilogy	15
11.4 Nonlinear ECM cointegration (<code>xtnonlincoint</code>)	15
12 Part XI — A worked empirical application (narrative)	16
13 Beginner FAQ	16
14 Exercises	17
15 One-page cheat-sheet	18
Key references	18

Notation and symbols

Symbol	Meaning
i, t	unit index $i = 1, \dots, N$; time index $t = 1, \dots, T$
N, T	number of cross-sectional units; number of time periods
$y_{it}, \mathbf{x}_{it}$	dependent variable; vector of K regressors for unit i at time t
α_i	individual (fixed) effect — unit-specific intercept
$\beta, \beta_i(t)$	slope vector; time-varying / unit-specific slope (breaks live here)
$\mathbf{f}_t, \gamma_i$	unobserved common factors; unit-specific factor loadings
e_{it}, ε_{it}	idiosyncratic errors
$T_B, \lambda = T_B/T$	break date; break fraction (relative position in the sample)
ω^2	long-run variance (LRV)
λ (penalty)	regularization tuning parameter (context makes it clear)
ρ_i	autoregressive / persistence coefficient of unit i
Δ	first-difference operator, $\Delta y_t = y_t - y_{t-1}$

1 Introduction: what this lecture is about

Modern applied panel work must survive *three problems at the same time*:

1. the series may be **non-stationary** (unit roots / stochastic trends);
2. the units may be **cross-sectionally dependent** (they share global shocks);
3. the relationships may contain **structural breaks** (they shift at some dates).

Tools that handle all three together are **third-generation** methods. This lecture builds them from zero: pre-concepts first, then the anatomy of a break, then cross-sectional dependence, then the three modern toolkits for modelling breaks (**dummies**, **Fourier**, **regularization**), and finally the full catalogue of tests and estimators — with **GAGFL**, **coefficient-by-coefficient (CBC)**, **Fourier CCEMG** and **Fourier panel ARDL** explained in detail, and Stata code for every step.

For beginners

No prior econometrics beyond a first regression course is assumed. Every symbol is explained the first time it appears. **Teal boxes** give the plain-English version; **purple boxes** are worked examples; **red boxes** flag common mistakes; **amber boxes** state the key idea; **cyan boxes** highlight the central equation.

2 Part I — Foundations from zero

2.1 The panel and its two dimensions

A *panel* (or *longitudinal*) dataset follows the same N units over T periods, giving NT observations. The basic linear panel model is

$$y_{it} = \alpha_i + \mathbf{x}_{it}'\beta + u_{it}, \quad i = 1, \dots, N; \quad t = 1, \dots, T, \quad (1)$$

where α_i (the **fixed effect**) absorbs all time-invariant, unit-specific heterogeneity. Two devices remove α_i :

- **Within transformation:** subtract each unit's time-mean, $\ddot{y}_{it} = y_{it} - \bar{y}_i$, killing α_i ;
- **First differencing:** $\Delta y_{it} = y_{it} - y_{i,t-1}$, also killing α_i .

Two asymptotic regimes — why they matter

- **Fixed- T , large- N :** $N \rightarrow \infty$, T small (micro panels; many macro panels). Break-date and factor consistency then come from having *many units*. Carrion et al. (2005), Kaddoura (2025), Baltagi–Feng–Kao work here.
- **Large- T :** $T \rightarrow \infty$ (alone or jointly with N). Needed for unit-root / cointegration limits built on *Brownian motion*. Westerlund (2006), KPY (2011), Banerjee–Carrion work here.

Using a fixed- T test on a short panel is fine; using a large- T test on $T = 8$ is not.

2.2 Stationarity, unit roots, and integration

A series is **stationary** (I(0)) if its mean and variance are constant and its autocovariances die out. It is **integrated of order one** (I(1)) if it must be differenced once to become I(0). The canonical I(1) process is the *random walk*

$$y_t = y_{t-1} + \varepsilon_t \iff \Delta y_t = \varepsilon_t, \quad \varepsilon_t \sim (0, \sigma^2), \quad (2)$$

for which $\text{Var}(y_t) = t\sigma^2$ grows without bound — shocks are *permanent*. In an AR(1) $y_t = \rho y_{t-1} + \varepsilon_t$, a **unit root** is $\rho = 1$; if $|\rho| < 1$ the series is mean-reverting and shocks are transitory.

Difference-stationary vs trend-stationary

Both an I(1) series and a trend-stationary series can drift upward. But the I(1) one has a *stochastic* trend (remove it by *differencing*); the trend-stationary one has a *deterministic* trend (remove it by *de-trending*). Mixing them up is the root of the **Perron critique** (Section 4.3): an unmodelled break makes a stationary series *look* like a unit root.

2.2.1 Two families of test: ADF vs KPSS

Family	H_0	Reject means	Panel-with-breaks
ADF / DF-type	unit root	stationarity	Im–Lee–Tieslau; CIPS; xtpqroot
KPSS / Hadri	stationarity	unit root	Carrion et al.; xtpkpss

The KPSS statistic uses *partial sums* of residuals:

The KPSS engine (drives xtpkpss)

$$\text{KPSS} = \frac{1}{T^2 \hat{\omega}^2} \sum_{t=1}^T S_t^2, \quad S_t = \sum_{s=1}^t \hat{\varepsilon}_s.$$

Under stationarity S_t stays bounded; under a unit root it wanders and inflates the statistic.

Long-run variance (LRV) and how it is estimated

The LRV is the spectral density at frequency zero,

$$\omega^2 = \gamma(0) + 2 \sum_{k=1}^{\infty} \gamma(k), \quad \gamma(k) = \text{Cov}(u_t, u_{t-k}),$$

estimated by a kernel (HAC) estimator with bandwidth ℓ : $\hat{\omega}^2 = \hat{\gamma}(0) + 2 \sum_{k=1}^{\ell} w(k/\ell) \hat{\gamma}(k)$, where $w(\cdot)$ is the **Bartlett** or **Quadratic-Spectral** kernel and ℓ the Andrews–Schwarz bandwidth $\lfloor 4(T/100)^{2/9} \rfloor$. The **homogeneous** version pools one ω^2 ; the **heterogeneous** version allows a

different ω_i^2 per unit (exactly the Z_{hom} vs Z_{het} split in **xtpkpss**).

2.3 Spurious regression and cointegration

The spurious-regression trap

Simulate two *independent* random walks y_t, x_t and regress y on x . You will typically get $R^2 > 0.7$ and a huge t -statistic — yet there is *no* relationship. The residuals are $I(1)$ (strongly autocorrelated), which is the tell-tale sign. **Lesson:** never run OLS on $I(1)$ levels without checking cointegration first.

Two $I(1)$ series are **cointegrated** if a linear combination of them is $I(0)$:

$$y_{it} = \alpha_i + \beta_i' \mathbf{x}_{it} + e_{it}, \quad \mathbf{x}_{it} \sim I(1), e_{it} \sim I(0) \Rightarrow \text{cointegration.} \quad (3)$$

Equivalently there is an **error-correction** form with adjustment speed $\phi_i < 0$:

$$\Delta y_{it} = \phi_i (y_{i,t-1} - \beta_i' \mathbf{x}_{i,t-1}) + (\text{short-run terms}) + \varepsilon_{it}. \quad (4)$$

A **structural break in cointegration** means α_i and/or β_i shift at some date — the equilibrium itself moves.

2.4 Why “generations”?

See the front-page figure. Each generation drops one more convenient assumption: **1st** assumes independence; **2nd** allows dependence but no breaks; **3rd** allows dependence *and* breaks. The rest of the lecture is the third column.

3 Part II — Anatomy of a structural break

3.1 Decomposition: where a break lives

The master decomposition

$$y_{it} = \underbrace{D_{it}}_{\text{deterministics (breaks here)}} + \underbrace{\gamma_i' \mathbf{f}_t}_{\text{common factor = dependence}} + \underbrace{e_{it}}_{\text{idiosyncratic}}.$$

Structural break

A change, at one or more dates T_B , in the parameters generating the data — the intercept, the trend, the slope, or the factor loadings.

3.2 Sharp breaks: level, trend, regime

Let $DU_t = \mathbf{1}\{t > T_B\}$ and $DT_t = (t - T_B)^+$. The three Perron-type sharp models are:

$$(A) \text{ level shift: } D_{it} = \mu_i + \theta_i DU_t, \quad (5)$$

$$(B) \text{ trend break: } D_{it} = \mu_i + \beta_i t + \delta_i DT_t, \quad (6)$$

$$(C) \text{ regime shift: } D_{it} = \mu_i + \theta_i DU_t + \beta_i t + \delta_i DT_t. \quad (7)$$

Reading a level shift

Suppose GDP growth averages 2% before 2008 and 1% after. Then $\mu_i = 2$, $\theta_i = -1$, $T_B = 2008$. A test that ignores $\theta_i DU_t$ sees a series that “fails to revert to 2%” and wrongly concludes a

unit root — the Perron critique in one line.

3.3 Where the break sits (four possibilities)

Break in	Meaning	Handled by
intercept	the mean level shifts	<code>xtpkps</code> , <code>xtpcointegwe</code>
slope $\beta(t)$	the marginal effect of $\mathbf{x}$ changes	<code>xtbreakmodel</code> , <code>xtcbc</code>
cointegrating vector	the long-run equilibrium shifts	<code>xtbreakcoint</code> , <code>xtcadfcoint</code>
factor loadings $\gamma_i(t)$	sensitivity to the common shock changes	<code>xtkpybreak</code>

3.4 Common vs heterogeneous breaks (across units)

Common / grouped / heterogeneous

Common: all units break at the same dates, $\beta_i(t) = \beta_j$ for $T_{j-1} < t \leq T_j$ with the same $T_1, \dots, T_m$ for every i . **Grouped:** units cluster into G latent groups, each group with its own dates. **Heterogeneous:** each unit has its own dates.

Key idea

Fewer distinct break schedules $\Rightarrow$ fewer parameters $\Rightarrow$ more power *if* the restriction is true. Common (1 schedule) $\subset$ Grouped (G schedules) $\subset$ Heterogeneous (N schedules). GAGFL estimates the grouped case; CCE-break methods assume common breaks; Fourier CCEMG absorbs heterogeneous-date breaks.

3.5 Vector vs coefficient-by-coefficient (across coefficients)

Type	Rule	Command
Vector break	all K coefficients break together at each T_B	<code>xtbreakmodel</code>
Coefficient-by-coefficient	each $\beta_k(t)$ breaks on its own schedule	<code>xtcbc</code>

Vector breaks over-split (they break stable coefficients too); CBC avoids this (Section 7.5).

3.6 Sharp vs smooth breaks

Sharp vs smooth

Sharp = step (dummy) at T_B . **Smooth** = gradual transition, modelled by

$$\text{logistic STR: } F(t) = [1 + \exp(-\gamma(t/T - \tau))]^{-1}, \quad (8)$$

$$\text{Fourier: } d_t = a_0 + \sum_k [a_k \sin \frac{2\pi kt}{T} + b_k \cos \frac{2\pi kt}{T}]. \quad (9)$$

γ sets sharpness ($\gamma \rightarrow \infty$ is a step), τ the location; a few Fourier frequencies k capture unknown, multiple, smooth breaks *with no date to estimate*.

3.7 A seven-question checklist

Answer these and the correct family is essentially pinned down:

1. Level, trend, or regime shift? 2. Intercept, slope, cointegrating vector, or loadings?
2. One or several breaks? 4. Known or estimated dates?

3. Common, grouped, or heterogeneous across units?
4. Vector or coefficient-by-coefficient?
7. Sharp or smooth?

Then always add: is there **cross-sectional dependence** and **non-stationarity**? (In this literature, almost always yes.)

4 Part III — Cross-sectional dependence (CSD)

4.1 What CSD is and why first-generation tests fail

Common-factor error structure

$$u_{it} = \gamma_i' \mathbf{f}_t + \varepsilon_{it}, \quad \mathbf{f}_t = (f_{1t}, \dots, f_{rt})' \text{ unobserved common factors.}$$

$\mathbf{f}_t$ is the shared shock (oil, crises, technology); γ_i is unit i 's exposure. The regressors load on the same factors, $\mathbf{x}_{it} = \Gamma_i' \mathbf{f}_t + \mathbf{v}_{it}$, which is exactly what makes the CCE proxy work.

Watch out

Ignore $\mathbf{f}_t$ and first-generation tests **over-reject** the unit-root null: the shared shock induces correlation across units that the “independent-panel” variance formula does not account for, so the pooled statistic is too large. You will “find” stationarity or cointegration that is an artefact of the common shock.

4.2 Two ways to remove CSD

4.2.1 (a) Cross-sectional averages — CCE (Pesaran, 2006)

Use $\bar{y}_t = \frac{1}{N} \sum_j y_{jt}$ and $\bar{\mathbf{x}}_t = \frac{1}{N} \sum_j \mathbf{x}_{jt}$ as proxies for $\mathbf{f}_t$ and partial them out. With $\bar{Z} = [\bar{y}, \bar{X}]$ and $\bar{M} = I - \bar{Z}(\bar{Z}'\bar{Z})^{-1}\bar{Z}'$,

CCE mean-group / pooled slope

$$\hat{\beta}_i = (X_i' \bar{M} X_i)^{-1} X_i' \bar{M} y_i, \quad \hat{\beta}_{\text{CCEMG}} = \frac{1}{N} \sum_i \hat{\beta}_i, \quad \hat{\beta}_{\text{CCEP}} = \left(\sum_i X_i' \bar{M} X_i \right)^{-1} \sum_i X_i' \bar{M} y_i.$$

Remarkable (KPY 2011): this stays valid even when $\mathbf{f}_t$ is I(1). That is the linchpin behind **xtkpybreak**.

4.2.2 (b) Common factors — PANIC / principal components (Bai–Ng, 2004)

Estimate the factors explicitly. Since levels mix integration orders, *difference first*, extract factors from differences by PCA, then re-cumulate:

$$\Delta y_{it} = \gamma_i' \Delta \mathbf{f}_t + \Delta e_{it} \xrightarrow{\text{PCA}} \hat{\mathbf{f}}_t = \sum_{s \leq t} \Delta \hat{\mathbf{f}}_s, \quad \hat{e}_{it} = \sum_{s \leq t} \Delta \hat{e}_{is}, \quad (10)$$

and test $\hat{\mathbf{f}}_t$ and $\hat{e}_{it}$ separately for unit roots. The number of factors r is chosen by the Bai–Ng criterion

$$\text{IC}_{p2}(k) = \ln(V(k, \hat{F}^k)) + k \left(\frac{N+T}{NT} \right) \ln(\min\{N, T\}), \quad (11)$$

minimised over k .

Averages vs factors, in one line

CCE = don't estimate the factors, subtract their footprint (cross-section averages). **PANIC** = estimate the factors by PCA. CCE is simpler; PANIC additionally tells you *whether* the non-stationarity is common or idiosyncratic.

4.3 The degree of dependence: classic binary vs modern**Classic vs modern view**

Classic: “is there dependence — yes or no?” Modern: *how strong*, indexed by the exponent α in $\sum_i |\gamma_i| = O(N^\alpha)$:

- $\alpha = 1$: **strong** (a genuine common factor). CCE/PANIC designed for this.
- $1/2 < \alpha < 1$: **semi-weak**. Banerjee–Carrion (2017) study CCE here.
- $\alpha \leq 1/2$: **weak** / **spatial** (local dependence). Spatial models / robust s.e.

4.4 Detecting CSD: the CD test**Pesaran CD test**

$$CD = \sqrt{\frac{2T}{N(N-1)}} \sum_{i < j} \hat{\rho}_{ij} \xrightarrow{d} N(0, 1), \quad \hat{\rho}_{ij} = \text{pairwise residual correlation.}$$

Reject $\Rightarrow$ dependence is present $\Rightarrow$ use 2nd/3rd-generation tools. Stata:

```
xtset id year
xtcd2 resid // Pesaran CD test on residuals (after a regression)
```

5 Part IV — Types of tests: classic vs modern**5.1 Classic: existence (yes / no)**

Classic tests answer a binary question — unit root yes/no, cointegration yes/no.

The null-direction trap

Half the cointegration tests have H_0 = cointegration (residual/LM: Westerlund 2006, `xtpcointegboot`) and half H_0 = no cointegration (ECM/factor: `xtpcointegwe`, `xtbreakcoint`). “Reject” means *opposite* things. State the null before the p-value.

5.2 Confirmatory analysis

Because unit-root and stationarity tests have low power, best practice is **confirmatory analysis**: run *both* an ADF-type (H_0 : unit root) and a KPSS-type (H_0 : stationarity). If they agree (one rejects, the other does not), the conclusion is robust; if both reject or neither rejects, the data are uninformative and you should be cautious.

5.3 Modern: the degree, not just the fact

Third-generation tests move from “exists / doesn't” to *how much*:

- **persistence across quantiles** — mean-reverting in calm times, persistent in the tails (CIPS(τ); `xtpqroot`);

- **strength of dependence** — weak / semi-weak / strong;
- **asymmetric adjustment** — faster correction on one side (nonlinear ECM; `xtnonlincoint`).

6 Part V — Three toolkits for modelling breaks

Toolkit	How it models the break	Best when
A. Dummies	step dummies at estimated dates (sharp)	discrete events, clear dates
B. Fourier	sine/cosine terms for smooth change	gradual / unknown-number breaks
C. Regularization	a penalty shrinking tiny changes to zero, keeping real breaks	many candidate dates, automatic selection

Toolkit A is Section 8 (Bai–Perron, Carrion KPSS, Westerlund). Toolkit B is Section 7 (Fourier). Toolkit C is Section 6 (regularization). We take C and B first because they are the modern, less familiar tools.

7 Part VI — Regularization explained, and GAGFL / PLS / CBC

7.1 What is regularization? (from zero)

The overfitting problem

Give a model as many free parameters as data points and it fits the sample perfectly but predicts terribly — it memorised noise. **Regularization** adds a *penalty* for complexity, trading a little bias for a big drop in variance.

The **bias–variance decomposition** of prediction error is

$$\mathbb{E}[(y - \hat{y})^2] = \underbrace{(\text{Bias})^2}_{\uparrow \text{ with } \lambda} + \underbrace{\text{Variance}}_{\downarrow \text{ with } \lambda} + \text{noise}, \quad (12)$$

so there is an optimal penalty λ^* that minimises the total. A regularized estimator solves

Fit + penalty

$$\min_{\beta} \underbrace{\sum_{i,t} (y_{it} - \mathbf{x}_{it}'\beta)^2}_{\text{fidelity}} + \lambda \cdot \underbrace{\text{pen}(\beta)}_{\text{complexity}}, \quad \lambda \geq 0.$$

7.1.1 Ridge, lasso, and why lasso hits zero

Penalty	Form	Effect
Ridge (ℓ_2)	$\lambda \sum_k \beta_k^2$	shrinks all coefficients, none exactly to zero
Lasso (ℓ_1)	$\lambda \sum_k \beta_k $	shrinks some coefficients <i>exactly</i> to zero (selection)
Fused lasso	$\lambda \sum_t \beta_t - \beta_{t-1} $	fuses consecutive periods into constant regimes

The lasso hits exactly zero because its constraint region $\sum |\beta_k| \leq c$ is a *diamond* with sharp corners on the axes; the elliptical fit contours touch a corner (a zero coefficient) with positive probability. The scalar solution is **soft-thresholding**:

$$\hat{\beta}^{\text{lasso}} = \text{sign}(\hat{\beta}^{\text{OLS}}) (|\hat{\beta}^{\text{OLS}}| - \lambda)^+. \quad (13)$$

7.1.2 Fused lasso = automatic break detection

Key idea

Penalise $|\beta_t - \beta_{t-1}|$. Most consecutive differences are shrunk to *exactly* zero (the coefficient is constant across those periods); the few non-zero differences are the **break dates**. So break detection becomes variable selection — no need to pre-specify the number of breaks.

For a *vector* coefficient we penalise the group norm $\|\beta_t - \beta_{t-1}\|$ (**group fused lasso**) and use **adaptive weights** for the oracle property:

Adaptive group fused lasso (the workhorse)

$$\min_{\{\beta_t\}} \frac{1}{N} \sum_{i,t} (y_{it} - \mathbf{x}'_{it} \beta_t)^2 + \lambda \sum_{t=2}^T w_t \|\beta_t - \beta_{t-1}\|, \quad w_t = \|\dot{\beta}_t - \dot{\beta}_{t-1}\|^{-\kappa}.$$

$\dot{\beta}_t$ are pilot estimates; κ (often 2) is the adaptive exponent; adaptive weighting gives the **oracle property** (asymptotically finds the true breaks as if known). The penalty λ is picked by a BIC-type information criterion.

7.1.3 How it is computed: block coordinate descent (BCD)

The objective is convex but non-smooth. **Block coordinate descent** updates one period's β_t at a time, holding the rest fixed, via a soft-thresholding / scalar line-search step; a full sweep $t = 1, \dots, T$ is repeated until the coefficient path stabilises. If the update norm falls below λw_{t-1} , the block is *fused* to the previous period ($\beta_t = \beta_{t-1}$).

7.2 PLS / AGFL — common breaks (Qian & Su, 2016)

The adaptive group fused lasso with a *single common* break schedule for all units. All K coefficients break together at dates selected by the fused penalty; λ by BIC. Command: `xtbreakmodel, method(pls)`.

7.3 GAGFL — grouped heterogeneous breaks (Okui & Wang, 2021)

GAGFL

Grouped Adaptive Group Fused Lasso: units belong to G latent groups; each group has its own break dates and regime coefficients; membership is unknown and estimated.

$$y_{it} = \alpha_i + \mathbf{x}'_{it} \beta_{g_i}(t) + \varepsilon_{it}, \quad g_i \in \{1, \dots, G\}.$$

The estimator iterates three stages to convergence:

Step 1. GFE initialisation. Cluster units into G groups (k-means style, Bonhomme–Manresa) on their time-varying coefficients, using many random starts to avoid local minima.

Step 2. Group-specific AGFL. Within each group g , solve the adaptive group fused lasso by BCD to detect that group's break dates and regime coefficients.

Step 3. Reassignment. Move each unit to the group whose estimated coefficient path minimises its residual sum of squares; return to Step 2.

G is chosen by BIC over a grid $G = 2, 3, \dots$. GAGFL nests common breaks ($G = 1$) and captures cluster-specific timing. Command: `xtbreakmodel, method(gagfl) groups(#)`.

7.4 CBC — coefficient-by-coefficient breaks (Kaddoura, 2025)

CBC / CBCL

Each coefficient $\beta_k(t)$ has its *own* number of breaks m_k and dates; stable coefficients are left unbroken.

Why it matters: the vector-break assumption splits *every* coefficient at the union of all break dates, even coefficients that never moved — inflating variance and masking the time-varying effects. CBC applies the fused penalty *per coefficient*:

CBC objective and its information criterion

$$L_\lambda(\beta) = \frac{1}{N} \sum_i \sum_{t \geq 2} (\tilde{y}_{it} - \bar{\mathbf{x}}'_{it} \beta)^2 + \lambda \sum_{t \geq 2} \sum_{k=1}^K w_{k,t} |\beta_{k,t} - \beta_{k,t-1}|,$$

$$\text{IC}_1(\lambda) = \hat{\sigma}^2(\lambda) + \phi \sum_k [\hat{m}_k(\lambda) + 1], \quad \phi = c \frac{\log N}{\sqrt{N}}.$$

Here $\tilde{y}_{it} = y_{it} - y_{i1}$ eliminates the fixed effect (first-period deviation), and $w_{k,t}$ are per-coefficient adaptive weights. **Sub-regime construction:** for coefficient k in its regime j , the other coefficients $\ell \neq k$ may break at *different* dates, so the estimator intersects regimes, $r_{\ell,c}^{(k,j)} = r_{k,j} \cap (\ell\text{'s } c\text{-th regime})$, and builds block-diagonal regressor matrices for the post-selection OLS. The option **csdemean** cross-section-demeans to absorb interactive fixed effects (Kaddoura–Westerlund 2023).

The headline result

In Kaddoura's US county-crime data, the *vector*-break estimator finds 2 breaks in *all* 16 coefficients; **CBC** finds breaks in only 3 controls, leaving the key deterrence coefficients unbroken. Command: **xtcbc**.

7.5 Breaks across quantiles (Zhang et al., 2022)

Quantile regression + adaptive fused lasso, pooled over quantiles, detects breaks that vary by quantile level (**xtquantilebreak**):

$$\min \sum_q \sum_{i,t} \rho_{\tau_q}(y_{it} - \alpha_i - \mathbf{x}'_{it} \beta_t(\tau_q)) + \lambda_1 \sum_i |\alpha_i| + \lambda_2 \sum_t w_t \|\beta_t - \beta_{t-1}\|, \quad (14)$$

where $\rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\})$ is the check function.

8 Part VII — Fourier methods in detail

8.1 The flexible Fourier form (FFF)

Flexible Fourier deterministic term

$$d_t = \alpha_0 + \alpha_1 t + \sum_{k=1}^n \left[a_k \sin\left(\frac{2\pi kt}{T}\right) + b_k \cos\left(\frac{2\pi kt}{T}\right) \right].$$

Gallant's theorem (1981): a few low-frequency sine/cosine terms approximate *any* smooth, bounded deviation, including several gradual breaks. Often a single frequency $n = 1$ suffices (Enders–Lee); the frequency is selected by minimising the SSR, and its significance is checked by an F -test on (a_k, b_k) . The **fractional frequency** version (Olayeni et al., 2021) searches non-integer $k \in [0.1, 5]$ for a sharper fit — the FFFFFF panel cointegration test.

8.2 Fourier CCEMG (Guliyev, 2026)

Key idea

When breaks occur at *different dates across units*, dummies explode in number. Fourier CCEMG augments the CCE regression with *unit-specific* Fourier terms: the cross-section averages filter the common factor while the Fourier terms *absorb the heterogeneously-timed breaks*.

Fourier CCEMG unit regression

$$y_{it} = \beta'_i \mathbf{x}_{it} + \underbrace{c_{0i} + c_{1i} \sin \frac{2\pi t}{T} + c_{2i} \cos \frac{2\pi t}{T}}_{\text{unit-specific smooth breaks}} + \underbrace{\delta'_i \bar{\mathbf{z}}_t}_{\text{CCE factor proxy}} + \varepsilon_{it}, \quad \hat{\beta}_{\text{F-CCEMG}} = \frac{1}{N} \sum_i \hat{\beta}_i.$$

with $\bar{\mathbf{z}}_t = [\bar{y}_t, \bar{\mathbf{x}}_t]$. In Monte Carlo it attains the lowest RMSE across weak/moderate/strong dependence with near-nominal coverage. (A sister estimator, **F-SURMG**, augments a seemingly-unrelated system with Fourier terms.)

8.3 Fourier panel ARDL / CS-ARDL

The **ARDL** model handles regressors of mixed order (I(0) and I(1)) and delivers long-run and short-run coefficients in one step. The Fourier-augmented panel ARDL lets the long-run relationship shift *smoothly*; the cross-sectionally augmented (CS-ARDL) version adds cross-section averages for dependence:

Fourier CS-ARDL (error-correction form)

$$\Delta y_{it} = \phi_i (y_{i,t-1} - \theta'_i \mathbf{x}_{i,t-1}) + \sum_j \psi'_{ij} \Delta \mathbf{x}_{i,t-j} + \underbrace{\lambda_{1i} \sin \frac{2\pi t}{T} + \lambda_{2i} \cos \frac{2\pi t}{T}}_{\text{smooth break}} + \underbrace{\xi'_i \bar{\mathbf{z}}_t}_{\text{CS augmentation}} + \varepsilon_{it}.$$

$\phi_i < 0$ is the **error-correction (adjustment) speed**; θ_i the long-run coefficients. Averaging the $\hat{\theta}_i, \hat{\phi}_i$ across units gives the **Mean-Group** estimator; restricting the long-run $\theta_i = \theta$ gives the **Pooled Mean Group (PMG)**. Command: **xtpfardl**.

9 Part VIII — The third-generation tests, catalogued

9.1 Panel unit-root / stationarity with breaks

9.1.1 Panel KPSS with multiple breaks — Carrion et al. (2005) — **xtpkpss**

Each unit's regression allows its own level/trend breaks; the panel averages and standardises:

Standardised panel KPSS

$$Z(\lambda) = \frac{\sqrt{N}(\overline{LM} - \bar{\xi})}{\sqrt{\bar{\varsigma}^2}} \xrightarrow{d} N(0, 1), \quad \overline{LM} = \frac{1}{N} \sum_i LM_i(\lambda_i).$$

$\bar{\xi}, \bar{\varsigma}^2$ are the mean/variance of the individual limit (they depend on the deterministic model and the break fractions). H_0 : stationarity; **right-tail** rejection. Homogeneous (Z_{hom}) and heterogeneous (Z_{het}) LRV; dates by Bai–Perron, number by LWZ, CSD by bootstrap.

9.1.2 PANIC with breaks — Bai & Carrion (2009)

Add breaks to D_{it} , then run PANIC: test the common factors (MQ tests for the number of stochastic trends) and the pooled idiosyncratic component ($\hat{P}_e$) separately for unit roots.

9.1.3 Im–Lee–Tieslau panel LM (2005)

A score (LM) unit-root test with the break *under the null*, so the statistic is **invariant** to the break size and location — estimated dates do not distort its $N(0, 1)$ limit.

9.1.4 Quantile CIPS and tFR — xtpqroot

CIPS(τ): a unit-root test at each quantile (asymmetric persistence). **tFR**: smooth (fractional Fourier) + sharp (LST) breaks together, CSD via sieve bootstrap.

$$Q_{\Delta y_{it}}(\tau \mid \mathcal{F}_{t-1}) = a_i(\tau) + \rho_i(\tau)y_{i,t-1} + d_i(\tau)\bar{y}_{t-1} + \cdots, \quad H_0 : \rho_i(\tau) = 1 \ \forall i, \tau.$$

9.2 Panel cointegration with breaks

Test	Command	H_0	CSD via
Westerlund (2006) LM	xtlmbreak	cointegration	bootstrap
Westerlund–Edgerton (2008)	xtpcointegwe	no cointegration	common factors
Westerlund–Edgerton (2007)	xtpcointegboot	cointegration	sieve bootstrap
Banerjee–Carrion (2015)	xtbreakcoint	no cointegration	factors
Banerjee–Carrion (2017)	xtcceoint	no cointegration	CCE
Banerjee–Carrion (2025)	xtcadfoint	no cointegration	CCE + factors
Nonlinear ECM (2017)	xtnonlincoint	no cointegration	sieve bootstrap

Westerlund (2006) is an LM (H_0 : cointegration) allowing multiple level/trend breaks at possibly different dates per unit. **Westerlund–Edgerton** give a factor-based (2008, PD_τ, PD_ϕ) and a bootstrap (LM^+ from FM-OLS, 2007) version. The **Banerjee–Carrion trilogy** progressively unifies breaks and dependence; the 2025 test allows breaks in the deterministic component, the cointegrating vector, *and* the factor loadings simultaneously. **Nonlinear ECM** adds asymmetric (regime-dependent) adjustment.

9.3 CCE with I(1) factors and breaks — xtkpybreak

The most fully third-generation *estimator*: CCE robust to I(1) factors (KPY 2011) *plus* multiple breaks in slopes K_0 *and* factor loadings K_1 , estimated jointly by the dynamic-programming Bai–Perron algorithm (Baltagi–Feng–Wang 2025):

Nonstationary panel with breaks in slopes and loadings

$$y_{it} = \mathbf{x}_{it}'\beta_i(K_0) + \gamma_i(K_1)'\mathbf{f}_t + \varepsilon_{it}, \quad \mathbf{f}_t = \mathbf{f}_{t-1} + \phi_t \text{ (I(1))}.$$

Empirical example: a common **1992 break** in international R&D spillovers, attributed to early-1990s globalisation.

9.4 Estimating and dating breaks: the shared machinery

9.4.1 Single break by SSR grid search

$$\hat{T}_B = \arg \min_{T_B \in [\varepsilon T, (1-\varepsilon)T]} \text{SSR}(T_B), \quad (15)$$

with **trimming** $\varepsilon \in [0.10, 0.15]$ keeping candidates away from the endpoints.

9.4.2 Multiple breaks

By **dynamic programming** (Bai–Perron 2003, global optimum in $O(T^2)$) or **sequential** detection. The number of breaks is chosen by a sequential $\sup F(\ell + 1 \mid \ell)$ test or an information criterion (BIC, LWZ).

9.4.3 Why the estimated date behaves as if known

The break fraction is **super-consistent**: $T(\hat{\lambda} - \lambda_0) = O_p(1)$, faster than $\sqrt{T}$, so downstream slopes and statistics have the same limit as if T_B were known.

9.4.4 Pooling to a standard normal

$$Z = \frac{\sqrt{N}(\bar{S} - \mu)}{\sqrt{\varsigma^2}} \xrightarrow{d} N(0, 1), \quad \bar{S} = \frac{1}{N} \sum_i S_i, \quad (16)$$

where (μ, ς^2) are the mean/variance of the individual limit — *they depend on the deterministic model and break count*, which is why every paper tabulates a different pair, and why you must declare the correct `model()` when you run the command.

10 Part IX — Practical workflow and Stata code

A defensible ordering for an empirical panel study (all commands from the `xt*` suite):

1. Test for CSD (Pesaran CD). 2. Test and date the breaks.
2. Unit-root / stationarity, break- and CSD-robust.
3. Number of common factors. 5. Cointegration with breaks.
4. Estimate the break-dated coefficients. 7. Interpret the long-run relation.

```
* ---- real data: Grunfeld investment panel (10 firms, 20 years) ----
webuse grunfeld, clear
xtset company year

* 1) cross-sectional dependence
xtcd2 invest // Pesaran CD test

* 3) unit-root / stationarity WITH breaks + CSD
xtpkps invest, model(constbreak) maxbreaks(2) graph // H0: stationary
xtpqroot invest, quantile(0.25 0.5 0.75) cdtest // quantile unit root
xtpqroot invest, fourier model(intercept) bootreps(299) // smooth+sharp breaks

* 5) cointegration WITH breaks + factors
xtpcointegwe invest mvalue, model(regimeshift) maxfactors(2) graph
xtbreakcoint invest mvalue kstock, model(levelshift) maxlag(1)
xtcadfcoint invest mvalue, model(3) breaks(1) nfactors(1) simulate(500)

* 6) estimate the break-dated coefficients
xtbreakmodel invest mvalue, method(pls) // common vector breaks
xtcbc invest mvalue kstock, graph // coefficient-by-coefficient
xtkpybreak break invest mvalue kstock, nbreaks(1) breakplot // CCE + I(1) + breaks
```

The one sentence to remember

A third-generation method = **remove the common factors** (CCE averages or PANIC principal components) + **let the deterministic / coefficient part break** (dummies, Fourier, or a fused lasso) + **use critical values that know the deterministic model**. Everything

else is which piece each author emphasises.

11 Part X — Each cointegration test in detail

11.1 Westerlund (2006) — LM test with multiple breaks (`xtlmbreak`)

An **LM** test whose null is *cointegration*. The cointegrating regression carries m_i breaks in level and trend, at possibly different dates per unit:

$$y_{it} = \alpha_i + \sum_k \theta_{ik} DU_{ikt} + \beta_i t + \sum_k \delta_{ik} DT_{ikt} + \gamma_i' \mathbf{x}_{it} + z_{it}, \quad z_{it} = \sum_{s \leq t} u_{is}.$$

The individual LM statistic is $LM_i = T^{-2} \sum_t S_{it}^2 / \hat{\omega}_i^2$ (partial sums of the residuals); the panel statistic $Z(M) = \sqrt{N}(\overline{LM} - \bar{\mu}) / \bar{\sigma} \rightarrow N(0, 1)$. **Right-tail** rejection $\Rightarrow$ no cointegration. Dates by SSR minimisation, CSD by bootstrap. *Use it* when you believe in the equilibrium and want to test whether a break destroys it.

11.2 Westerlund–Edgerton — factor and bootstrap versions

11.2.1 2008, factor-based (`xtpointegwe`)

Individual ADF regressions on the residuals, augmented with break dummies; CSD via a common-factor structure (PCA, number of factors by Bai–Ng ICp1). Two statistics PD_τ (t-ratio based) and PD_ϕ (coefficient based), both $\rightarrow N(0, 1)$. H_0 : no cointegration; **left-tail** rejection. Models: `nobreak` / `levelshift` / `regimeshift`.

11.2.2 2007, bootstrap (`xtpointegboot`)

The McCoskey–Kao LM^+ statistic from **FM-OLS** residuals (which correct endogeneity and serial correlation). Because asymptotic critical values are poorly sized under CSD, a **sieve bootstrap** (fit a VAR to the differences, resample innovations) supplies the critical values. H_0 : cointegration; right-tail rejection.

11.3 The Banerjee–Carrion trilogy

Paper	What it adds	Command
2015 (JAE)	cointegration test valid under breaks and CSD (factors)	<code>xtbreakcoint</code>
2017 (JTSA)	CCE-based test; consistent long-run parameter under weak/semi-weak/strong dependence	<code>xtcceoint</code>
2025 (JBES)	breaks in the deterministic part, the cointegrating vector, AND the factor loadings	<code>xtcadfcoint</code>

The 2025 model (most general) is

$$y_{it} = D_{it}(\text{breaks}) + \theta_i' \mathbf{x}_{it}(\text{breaks}) + \gamma_i(\text{breaks})' \mathbf{f}_t + e_{it}, \quad H_0 : e_{it} \sim I(1) \text{ (no coint.)}.$$

A defactored CADF-type statistic on the residuals; cross-section averages remove $\mathbf{f}_t$; break dates by SSR. **Left-tail** rejection $\Rightarrow$ cointegration.

11.4 Nonlinear ECM cointegration (`xtnonlincoint`)

The first **nonlinear** panel cointegration test: adjustment toward equilibrium is *asymmetric*, modelled by a logistic transition so that the correction speed differs above vs below the equilibrium; a modified Wald statistic tests the ECM term; CSD via sieve bootstrap:

$$\Delta y_{it} = \phi_i [1 + \exp(-\gamma_i(e_{i,t-1} - c_i))]^{-1} e_{i,t-1} + (\text{short-run}) + \varepsilon_{it}, \quad H_0 : \phi_i = 0 \forall i.$$

12 Part XI — A worked empirical application (narrative)

Suppose we study the **energy–growth** relationship for $N = 20$ countries over $T = 40$ years, with $y = \log$ GDP per capita and $x = \log$ energy use per capita. Here is the full third-generation workflow, with how to *read* each step.

Step 1 — cross-sectional dependence

Run `xtcd2`. Suppose $CD = 18.4$ ($p < 0.001$). **Read:** strong dependence — global shocks (oil prices, crises) hit all countries. First-generation tests are now invalid; we must use CCE / factor / bootstrap methods.

Step 2 — date the breaks

Run `xtgets ...`, `fesis`. Suppose it flags breaks clustered around 1979 and 2008. **Read:** candidate regime changes (oil shock; global financial crisis) — we will let the tests estimate these rather than impose them.

Step 3 — unit root with breaks

Run `xtpkpss y, model(trendbreak) maxbreaks(2)`. Suppose $Z_{\text{het}} = 4.1$ ($p < 0.01$). **Read:** reject stationarity $\Rightarrow y$ is $I(1)$ even after allowing breaks. Repeat for x : also $I(1)$. Both series are integrated, so a *levels* regression risks being spurious — we need cointegration.

Step 4 — cointegration with breaks

Run `xtpcointegwe y x, model(regimeshift) maxfactors(2)`. Suppose $PD_{\tau} = -3.7$ ($p < 0.01$). **Read:** reject H_0 of no cointegration $\Rightarrow$ a long-run energy–growth equilibrium exists, and it *shifts* at the estimated date. The relationship is real, not spurious.

Step 5 — estimate the break-dated relationship

Run `xtkpybreak break y x, nbreaks(1)`. Suppose the long-run elasticity is 0.65 before the break and 0.42 after. **Read:** energy became *less* tightly linked to growth after the break (efficiency gains / structural change toward services). Match the estimated date to the real event.

Interpretation discipline

Always report: (i) the *null* of each test, (ii) the *estimated break dates with a confidence interval*, and (iii) the *economic event* you attribute each break to. A break with no story is a red flag for a spurious artefact.

13 Beginner FAQ

Q. Do I difference before or after removing the factors? For PANIC, *difference first*, then extract factors, then re-cumulate. For CCE you do not difference — you add the cross-section averages to the levels regression.

Q. My panel is unbalanced. Can I still use these? Most break commands need a *balanced* panel. Balance it first (`tsfill`, imputation) or use a missing-value-robust unit-root test as a pre-step.

Q. How many breaks should I allow? Set a sensible `maxbreaks()` (2–3 for short T) and let the information criterion (LWZ/BIC) choose. Too many candidate breaks on a short T over-fits.

Q. Dummy, Fourier, or regularization? Dummies for *discrete, known-ish* events; Fourier for *gradual / unknown-number* breaks; regularization when you want the method to *select* the breaks automatically among many candidates.

Q. Vector break or CBC? If you suspect only *some* coefficients move, use CBC (`xtcbc`); it avoids splitting the stable ones and keeps their standard errors small.

Q. What does “super-consistency” buy me? It means the estimated break date converges so fast that the downstream slope estimates behave *as if the date were known* — so you can plug in $\hat{T}_B$ without an extra penalty.

14 Exercises

1. **Concept.** Explain, in one sentence each, the difference between (a) common vs heterogeneous breaks, (b) vector vs coefficient-by-coefficient breaks, (c) sharp vs smooth breaks.
2. **Null direction.** For each of `xtpkps`, `xtpointegwe`, `xtpointegboot`, state H_0 and what a rejection implies.
3. **Regularization.** Starting from the fused-lasso penalty $\lambda \sum_t |\beta_t - \beta_{t-1}|$, explain why increasing λ reduces the number of detected breaks. What selects λ ?
4. **CCE vs PANIC.** Give one situation where you would prefer CCE and one where you would prefer PANIC, and justify.
5. **Fourier.** Why can a single Fourier frequency approximate several gradual breaks, and why is that impossible with a single dummy?
6. **Applied (Stata).** On `webuse grunfeld`, run `xtcbc invest mvalue kstock, graph`. Which coefficients break and which stay constant? Compare with `xtbreakmodel ..., method(pls)` and explain the difference in the number of breaks.
7. **Workflow.** Write the seven-step workflow for a panel where the CD test rejects and a break is suspected in the long-run slope. Name a command for each step.

15 One-page cheat-sheet

Question	Method	Command
Is there dependence?	Pesaran CD	xtcd2
Detect / date breaks	indicator saturation	xtgets
Unit root with breaks?	panel KPSS / quantile CIPS / tFR	xtpkpss, xtpqroot
Cointegration with breaks?	Westerlund; Banerjee–Carrion	xtlmbreak, xtpcointegwe, xtbreakcoint, xtcceoint, xtcadfcoint
Bootstrap cointegration	FM-OLS LM^+	xtpcointegboot
Nonlinear / asymmetric ECM	logistic transition	xtnonlincoint
Estimate breaks (common)	Qian–Su / BFK / SaRa	xtbreakmodel
Estimate breaks (grouped)	Okui–Wang GAGFL	xtbreakmodel, method(gagfl)
Estimate breaks (per-coef)	Kaddoura CBC	xtcbc
CCE + I(1) factors + breaks	KPY / Baltagi–Feng–Wang	xtkpybreak
Breaks across quantiles	shrinkage QR	xtquantilebreak
Endogenous regressor + break	Baltagi–Feng–Kao	xtbfkbreak
Dynamic panel, breaks + CSD	robust dynamic estimators	xtdynestimb
Panel VAR + cointegration + breaks	panel VECM	xtpvarcoint

Key references

- Bai, J. & Ng, S. (2002, 2004). Determining the number of factors; A PANIC attack on unit roots and cointegration. *Econometrica* 70(1); 72(4).
- Bai, J. & Carrion-i-Silvestre, J.L. (2009). Structural changes, common stochastic trends, and unit roots in panel data. *Review of Economic Studies* 76(2).
- Bai, J. & Perron, P. (1998, 2003). Estimating and testing linear models with multiple structural changes. *Econometrica* 66(1); *JAE* 18(1).
- Baltagi, B.H., Feng, Q. & Kao, C. (2016, 2019). Estimation of heterogeneous panels with structural breaks. *Journal of Econometrics* 191(1).
- Baltagi, B.H., Feng, Q. & Wang, W. (2025). Nonstationary heterogeneous panels with multiple structural changes. *Econometric Reviews*.
- Banerjee, A. & Carrion-i-Silvestre, J.L. (2015, 2017, 2025). Panel cointegration with breaks and cross-section dependence. *JAE*; *JTSA*; *JBES*.
- Bonhomme, S. & Manresa, E. (2015). Grouped patterns of heterogeneity in panel data. *Econometrica* 83(3).
- Carrion-i-Silvestre, J.L., del Barrio-Castro, T. & López-Bazo, E. (2005). Breaking the panels. *Econometrics Journal* 8(2).

- Corakci, A. & Omay, T. (2023). Panel unit root test with smooth and sharp breaks. *Renewable Energy* 205.
- Enders, W. & Lee, J. (2012). The flexible Fourier form and unit-root tests. *Economics Letters* 117(1).
- Kaddoura, Y. (2025). Estimating coefficient-by-coefficient breaks in panel data models. *Journal of Econometrics* 249.
- Kaddoura, Y. & Westerlund, J. (2023). Panel models with random interactive effects and multiple breaks when T is fixed. *JBES* 41(3).
- Kapetanios, G., Pesaran, M.H. & Yamagata, T. (2011). Panels with non-stationary multifactor error structures. *Journal of Econometrics* 160(2).
- Okui, R. & Wang, W. (2021). Heterogeneous structural breaks in panel data models. *Journal of Econometrics* 220(2).
- Olayeni, R.O., Tiwari, A.K. & Wohar, M.E. (2021). Fractional frequency flexible Fourier form for panel cointegration. *Applied Economics Letters* 28(6).
- Pesaran, M.H. (2006, 2007). CCE estimation; a simple panel unit-root test. *Econometrica; JAE*.
- Qian, J. & Su, L. (2016). Shrinkage estimation of common breaks via adaptive group fused lasso. *Journal of Econometrics* 191(1).
- Westerlund, J. (2006, 2007); Westerlund, J. & Edgerton, D.L. (2007, 2008). Panel cointegration with structural breaks. *OBES; Economics Letters*.
- Zhang, L., Zhu, Z., Feng, X. & He, Y. (2022). Shrinkage quantile regression for panel data with multiple structural breaks. *Canadian Journal of Statistics* 50(3).

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